

DAAKYE

VENDOR SPACE

Mobile Platform



Comprehensive Project Write-Up, System Design Overview & Implementation Roadmap

Digitizing informal retail for Ghanaian vendors

Introduction

- Mobile-first inventory and sales management system for small and medium-scale vendors in Ghanaian local markets.
- Digitizes daily operations currently handled via handwritten notebooks, manual stock counts, and paper-based records.
- Lightweight, affordable, scalable — optimized for low-end Android devices and low-connectivity environments.
- Enables vendors to manage inventory, record sales, track revenue, receive alerts, and analyze business performance.

KEY FOCUS

- Usability
- Affordability
- Low data use
- Offline support

The Challenge

A large percentage of vendors in Ghanaian local markets operate without structured digital systems — creating recurring operational challenges:

Frequent stockouts

Overstocking of slow-moving goods

Inaccurate sales records

Time-consuming reconciliation

Human arithmetic errors

Poor revenue tracking

No business insights

Difficulty accessing loans

EXISTING SYSTEMS FALL SHORT BECAUSE:

Too expensive

Overly complex

Internet-dependent

Not optimized for
African markets

Daakye Vendor Space Platform

A modular, mobile-first inventory and sales management system that digitizes vendor operations while remaining simple, scalable, and accessible.



Real-time Inventory Tracking



Sales Recording



Auto Stock Deductions



Low-stock Alerts



Business Analytics



Transaction History

Who We Serve & What We Aim to Achieve

TARGET USERS

- Market vendors
- Kiosk operators
- Mini marts
- Small retail shops
- Wholesalers
- Local store attendants

Low digital literacy • Unstable internet • Low-cost Android devices

PROJECT OBJECTIVES

- Digitize inventory management
- Simplify sales recording
- Automate stock deductions
- Improve business decision-making via analytics
- Support offline-capable operations
- Generate transaction histories for financial inclusion

Modular Monolith Architecture



React Native Mobile App

TypeScript · Expo



REST API Layer

Node.js · Express.js



Business Modules

Auth · Inventory · Sales · Analytics



MongoDB Atlas

Cloud Database

Simplified deployment

Reduced infrastructure complexity

Supports future microservices migration

Tech Stack



MOBILE

React Native · Expo · TypeScript · React Navigation · Axios · AsyncStorage



BACKEND

Node.js · Express.js · JWT Authentication · Bcrypt



DATABASE & SERVICES

MongoDB Atlas · Firebase Cloud Messaging · Victory Native (Charts)



DEV TOOLS

Visual Studio Code · GitHub · Postman · Figma

Database Design & API Structure

DATABASE COLLECTIONS

Users	<i>id, name, phone, email, password, role</i>
Products	<i>id, name, category, price, quantity, lowStockThreshold</i>
Sales	<i>id, products, totalAmount, date, soldBy</i>
Notifications	<i>id, message, productId, status, createdAt</i>

API ENDPOINTS

Auth	POST /auth/register POST /auth/login
Inventory	GET /products POST /products PUT /products/:id DELETE /products/:id
Sales	POST /sales GET /sales PUT /sales/:id DELETE /sales/:id
Analytics	GET /analytics/revenue GET /analytics/top-products

7-Phase Roadmap



01

**Research &
Planning**

*Requirements, workflow
mapping, feature
prioritization*

02

UI/UX Design

*Wireframes, mobile layout
navigation flow*

03

Backend Development

*Auth, Inventory, Sales APIs &
Notification logic*

04

**Mobile
Development**

*React Native setup, screen
API integration*

05

**Analytics
&
Notifs**

*Charts, revenue metrics, push
notifications*

06

**Testing
&
Optimization**

*Functional & device testing,
bug fixing*

07

Pilot Deployment

*Real-user testing, vendor
feedback, UX refinement*

Est. 10 Weeks to MVP

Timeline & Budget Estimate

TIMELINE TO MVP

Planning & Design	2 weeks
Backend Development	2 weeks
Mobile Development	2 weeks
Testing & Optimization	2 weeks
Pilot Testing	2 weeks

Total: ~10 Weeks

ESTIMATED BUDGET

Development Tools	Free
Hosting	Free tier
Database (Atlas)	Free tier
Firebase Notifications	Free
GitHub / Expo	Free
Google Play Console	~\$25 one-time
Domain Name	~\$10–20/year

MVP Budget: ~\$0 – \$50

What's Next



Offline-first Sync



Barcode Scanning



Mobile Money Integration



AI Stock Forecasting



Voice-assisted Sales



Local Language Support



Supplier Management



Multi-branch Management



WhatsApp Receipt Integration

Conclusion

Daakye Vendor Space bridges the digital management gap in Ghana's informal retail sector — delivering a simple, scalable, and mobile-first solution that empowers vendors with real digital tools.

Mobile-first Design

Modular Architecture

Offline-capable
Planning

Intuitive Workflows

Scalable System Design

● Inventory Management

● Sales Tracking

● Business Analytics

● Financial Inclusion

A strong foundation for future expansion into fintech, intelligent analytics, and African digital commerce infrastructure.